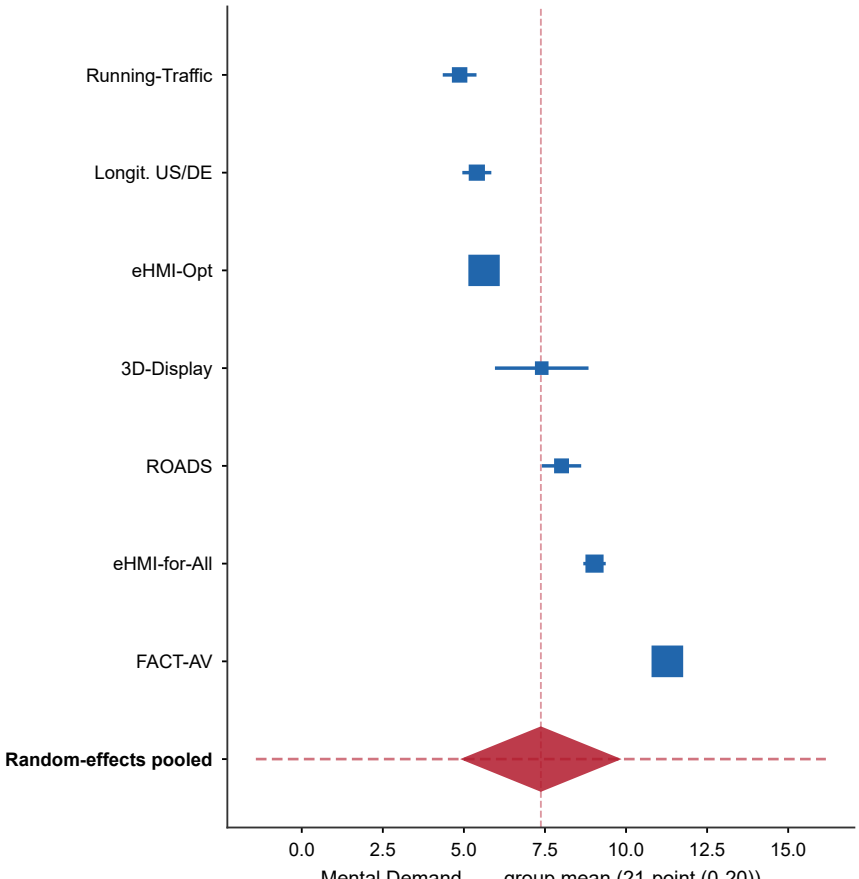
## Evidence synthesis unlocked by standardization: NASA-TLX Mental Demand

Pooling is only possible because heterogeneous source labels were mapped to one canonical DV. Very high I² shows that shared names still mask scale/population differences.




| | N obs | Mean | Wt | 95% CI |
|---|---|---|---|---|
| Running-Traffic | 288 | 4.87 | 4.4% | [4.3, 5.4] |
| Longit. US/DE | 296 | 5.40 | 6.0% | [5.0, 5.8] |
| eHMI-Opt | 1,480 | 5.62 | 37.6% | [5.4, 5.8] |
| 3D-Display | 45 | 7.40 | 0.6% | [6.0, 8.8] |
| ROADS | 276 | 8.01 | 3.3% | [7.4, 8.6] |
| eHMI-for-All | 720 | 9.03 | 10.0% | [8.7, 9.4] |
| FACT-AV | 2,600 | 11.27 | 38.1% | [11.1, 11.5] |
| **Random-effects pooled** | **5,705** | **7.37** | **100%** | **[5.0, 9.8]** |

Random-effects (DerSimonian–Laird), k = 7 studies. Heterogeneity: $I^2$ = 99.7%, $\tau^2$ = 10.22, Q(6) = 2287, p < 0.001
Hartung–Knapp–Sidik–Jonkman 95% CI (t, 6 df): [5.2, 9.5]